

Course: Networked Control

Examination procedure

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General description

The Exam consists of **two parts**.

- Regarding the course **Part II (Decentralized and distributed control structures)** a *Project Work* will have to be completed and presented in an oral discussion IN PRESENCE. The lecture given on 02/11/2021 was dedicated to the *Project Work*.
- Regarding the course **Part III (Network control systems)**, a *Written Test* will be taken.

While the written test can be taken only at the “official” Exam dates (e.g., 12/01/2022 and 01/02/2022), the Project work can be presented at any date (either before or after taking the Written Test), under request by e-mail. **To take the Written Test, students must be enrolled.**

- If, at the date of the Written Test, students have already presented the Project Work, a mark (average of the marks obtained in the two parts) will be assigned and registered.
- If, at the date of the Written Test, students have still to present the Project Work, a “partial” mark will be assigned. To register the final mark, students have to take both parts: note that registration can be made only in correspondence with the “official exam sessions”.

Written Test

Written test

- It focuses on **Part III** (Networked Control Systems).
- It consists of
 - **numerical exercises** of the types proposed during the Exercise lectures (23/09/2021, 18/11/2021, 9/12/2021);
 - **questions**, e.g., regarding
 - **Control networks and data networks**;
 - **delays in control networks** (and more specifically in the source node, on the network channel, and in the destination node, including the definition of frame time, propagation time, etc.);
 - the main **features of network protocols**;
 - the statement of the **main results in control under quantization**;
 - the **stability of a networked control system under delays** and maximal allowable transfer interval;
 - packet dropout rate (definitions) and stability under **packet dropouts**.

Written test

- The test is conceived to be feasible in 1h 30 m, but 2h will be granted to allow all (prepared) students to complete the test.
- Past tests with solutions are available on WeBEEP.
- Future written tests will be delivered in presence.
- A pdf document with some equations will be made available during the test.

Written test - available equations

Control under discretization

Consider the continuous-time scalar system

$$\dot{x} = ax + bu$$

where, for simplicity, we assume that $b > 0$.

The dynamics of the system under control is

$$x_{k+1} = fx_k + gu_k$$

If $a > 0$ (open-loop unstable system) the control law defined in case we operate at the rate limits is

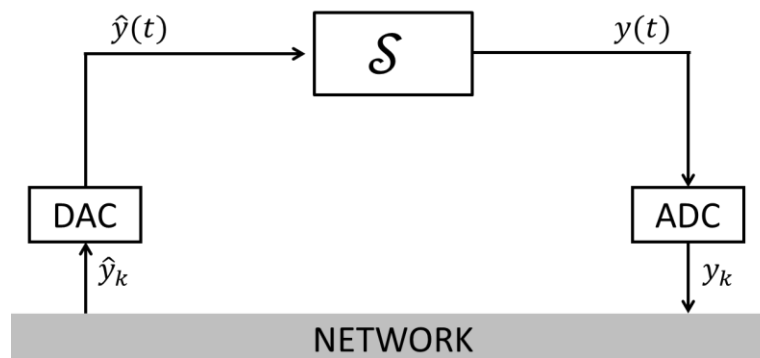
$$u_k = \begin{cases} u_{\max} & x_k \leq \frac{b}{a} \left(-u_{\max} + \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\max} - i \frac{u_{\max} - u_{\min}}{N - 1} & \frac{b}{a} \left(-u_{\max} + i \frac{u_{\max} - u_{\min}}{N} \right) < x_k \leq \frac{b}{a} \left(-u_{\max} + (i + 1) \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\min} & x_k > \frac{b}{a} \left(-u_{\min} + \frac{u_{\min} - u_{\max}}{N} \right) \end{cases}$$

→ The set $\left[-\frac{b}{a} u_{\max}, -\frac{b}{a} u_{\min} \right]$ is invariant.

Written test - available equations

Delays and packet dropouts

Consider the following scheme



where the dynamics of $\mathcal{S}$ is

$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

In case sampling time h_k is used at each time instant and the communication is subject to a delay τ_k , the discrete-time state x_k evolves according to

$$x_{k+1} = e^{Ah_k}x_k + \Gamma(h_k - \tau_k)B\hat{y}_k + e^{A(h_k - \tau_k)}\Gamma(\tau_k)B\hat{y}_{k-1}$$

where $\Gamma(\mu) = \int_0^\mu e^{A\nu} d\nu$.

- When transmission takes place (**no dropouts**), $\hat{y}_k = Cx_k$.
- When there is a **dropout** we set $\hat{y}_k = \hat{y}_{k-1}$